

DIS Department of Mathematics
MYP Grade 10 Extended - Unit Test Trigonometry

Name: Answers

Date: _____

You will be assessed against criteria A and D. You may use your GDC for all questions.
Show all your working!

The general sine function is

$$y = a \sin(b(x - c)) + d \quad \text{where } b > 0.$$

affects
affects
affects
affects
amplitude **period** **horizontal translation** **vertical translation**

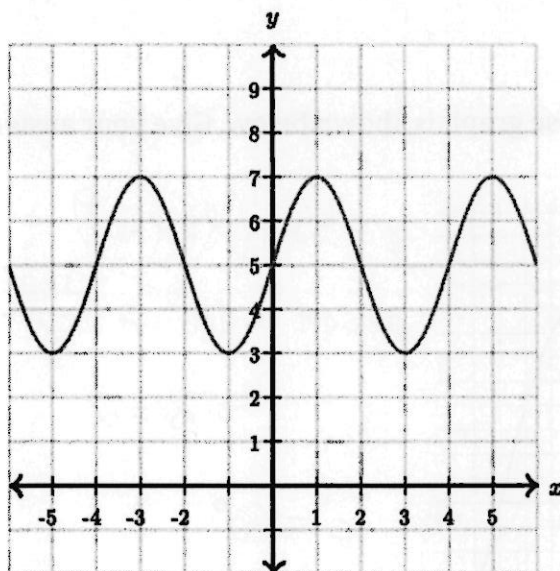
The **principal axis** of the general sine function is $y = d$.

The **period** of the general sine function is $\frac{2\pi}{b}$ or $\frac{360^\circ}{b}$.

The **amplitude** of the general sine function is $|a|$.

Criterion A Questions

1. Consider the diagram below, which shows part of the graph of a sine function.



- (a) Write down (i) the equation of the principal axis: $y = 5$
(ii) the amplitude: 2
(iii) the period (= wavelength): 4

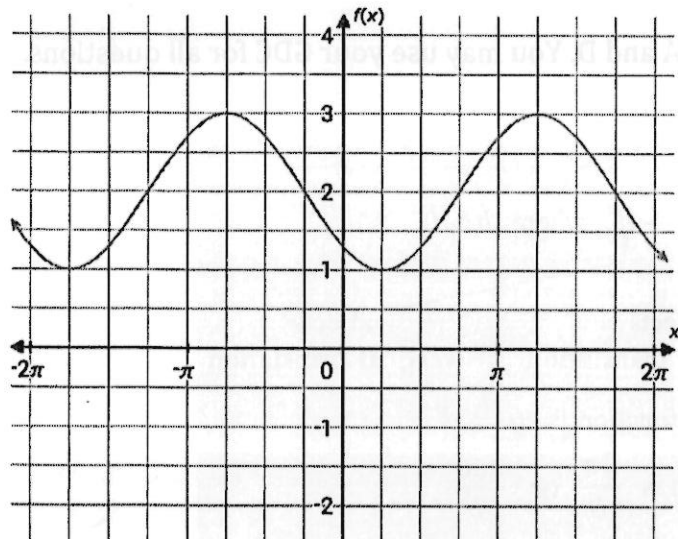
- (b) Find an equation of the sine function, giving your answer in the form $y = A \sin(Bx + C) + D$.

$$y = 2 \sin\left(\frac{\pi}{2}(x - c)\right) + 5 \quad c = 0$$

$$\underline{\underline{y = 2 \sin\left(\frac{\pi}{2}x\right) + 5}}$$

$$4 = \frac{2\pi}{b} \rightarrow \underline{\underline{b = \frac{\pi}{2}}}$$

2. Find an equation for the sine function whose graph is shown below. Give your answer in the form $y = A\sin(Bx + C) + D$.



$$y = \sin(x - C) + 2$$

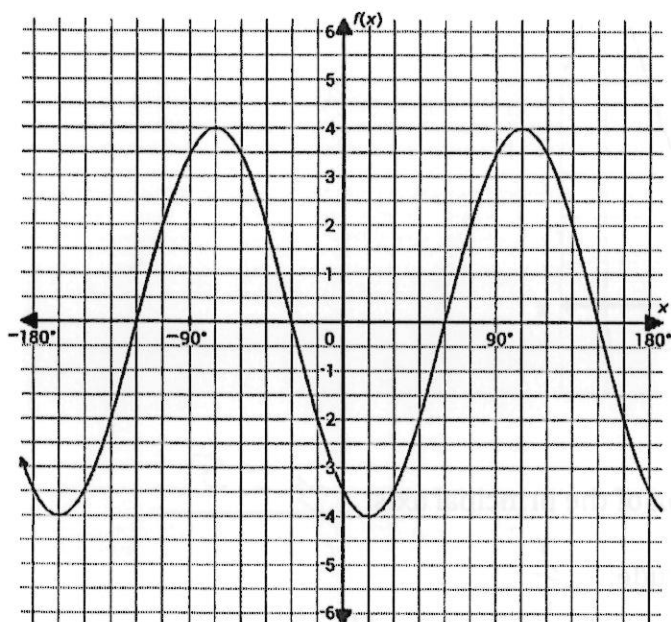
$$2\pi = \frac{2\pi}{b} \rightarrow b = 1$$

$$C = -\frac{5\pi}{4} \quad (\text{or } 2.36)$$

$$\therefore y = \sin\left(x + \frac{5\pi}{4}\right) + 2$$

$$\text{or } y = \sin(x - 2.36) + 2$$

3. Find an equation for the function whose graph is shown below. Give your answer in the form $y = A\sin(Bx + C) + D$.



$$y = 4\sin(2(x - c))$$

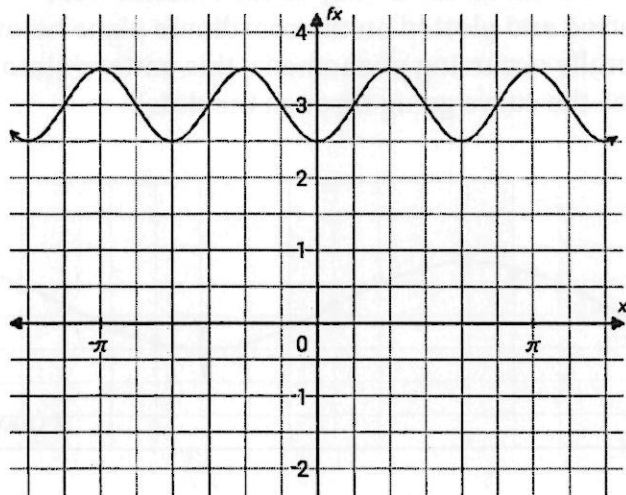
$$\text{period} = 180^\circ = \frac{360^\circ}{b}$$

$$\rightarrow b = 2$$

$$c = 60^\circ$$

$$y = 4\sin(2x - 120^\circ)$$

4. Find an equation for the function whose graph is shown below. Give your answer in the form $y = A \sin(Bx + C) + D$.



$$y = \frac{1}{2} \sin(3(x - c)) + 3$$

$$\text{period} = \frac{2\pi}{3} \rightarrow b = 3$$

$$c = \frac{5\pi}{6}$$

$$y = \frac{1}{2} \sin\left(3x - \frac{5\pi}{2}\right) + 3$$

5. A function has equation $y = -2 \sin(3x + 4) - 5$. $= -2 \sin\left(3\left(x - \frac{5}{3}\right)\right) - 5$

Write down the: (a) amplitude: 2

(b) equation of the principal axis: $y = -5$

(c) period: $\frac{2\pi}{3}$

$$\text{period} = \frac{2\pi}{3}$$

6. A sine function has principal axis $y = \pi$, period π , and one of its maximum points has coordinates $\left(\frac{\pi}{4}, 2\pi\right)$. Find the equation of this function, giving your answer in the form $y = A \sin(Bx + C) + D$.

$$y = \pi \sin(2(x - c)) + \pi$$

$$\text{amplitude} = \pi$$

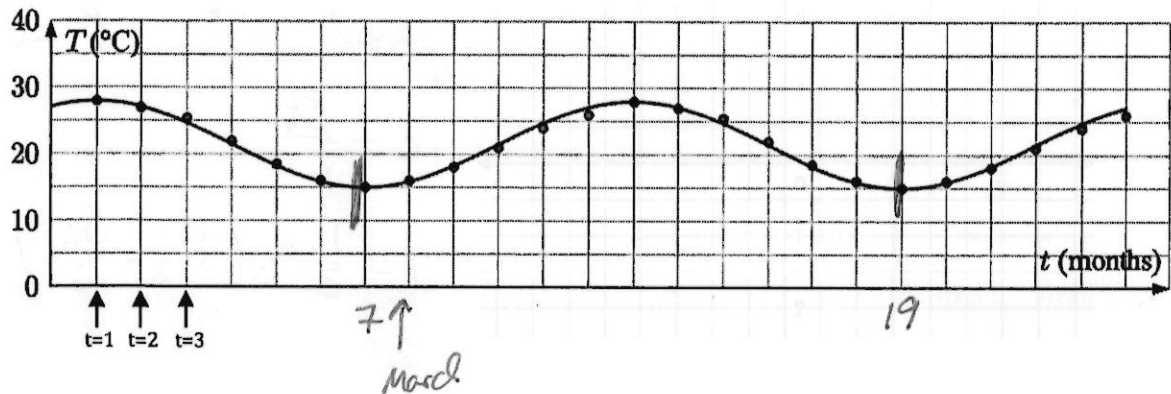
$$\text{period} = \pi = \frac{2\pi}{b} \rightarrow b = 2$$

$$\left(\frac{\pi}{4}, 2\pi\right): 2\pi = \pi \sin\left(2\left(\frac{\pi}{4} - c\right)\right) + \pi \rightarrow c = 0$$

$$\therefore \underline{y = \pi \sin(2x) + \pi}$$

Criterion D Questions

1. The temperatures in Juttastadt at 11:30 am on the 1st day of each month were measured over a 24 month time period and plotted on the coordinate plane below (see the dots). Just like many other naturally occurring phenomena, this pattern almost perfectly follows a sine function (see the curve going through the dots).



$t = 1$ means August 2022, $t = 2$ means September 2022, $t = 3$ means October 2022 and so on.

- (a) Write down the period of this function, including the correct unit.

12 months

- (b) The two minima occur at (7, 15°C) and (19, 15°C). Given that the amplitude is 6.75°C, what is the equation of the principal axis?

$$T = 21.75 \text{ (}^\circ\text{C)}$$

- (c) Using radians, find an equation of the function that models the temperature T in Juttastadt over time. Give your answer in the form $T(t) = A \sin(Bt + C) + D$.

$$T = 6.75 \sin\left(\frac{\pi}{6}(t - C)\right) + 21.75$$

$$\frac{2\pi}{B} = 12 \rightarrow B = \frac{\pi}{6}, \quad C = 10$$

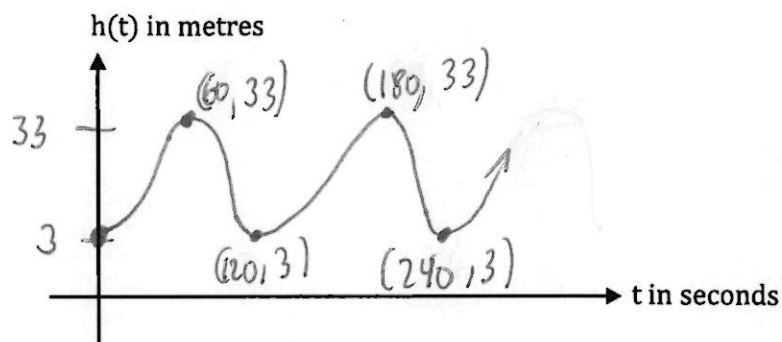
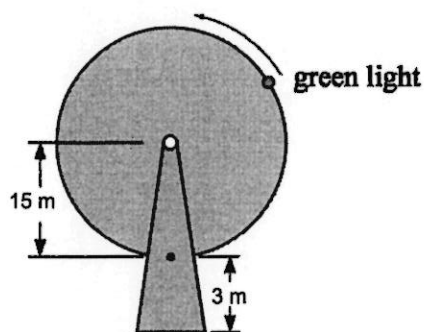
$$\rightarrow T(t) = 6.75 \sin\left(\frac{\pi}{6}t - \frac{5\pi}{3}\right) + 21.75$$

- (d) Use your equation from (c) to estimate the temperature in Juttastadt at 11:30 am on March 15th.

$$T(8.5) = 17.0 \text{ }^\circ\text{C}$$

2. A Ferris wheel rotates at a constant speed of half a rotation per minute. The wheel's radius is 15 m and the bottom of the wheel is 3 m above ground level. An engineer needs to model the height of a particular gondola (marked by a green light on the bottom of it) over time. His goal is to find a formula that calculates the height of the green light above ground level (h) at any given time (t).

For $t = 0$ s, the green light is at its lowest point (see the diagram below).



- (a) The function $h(t)$ follows a sine curve. Sketch it on the set of axes above, indicating the coordinates of any minima or maxima.
- (b) From your sketch, determine (i) the equation of the principal axis, (ii) the period, and (iii) the amplitude.

(i) $h = 18$ (m) (ii) 15 (m)

(ii) period = 2 (min)
= 120 (sec)

- (c) Using radians, find a function that calculates the height of the green light above ground level h (in m) at any given time t (in seconds). Give your answer in the form $h(t) = A \sin(Bt + C) + D$.

using minutes

$$\begin{cases} h(t) = 15 \sin(\pi(t - c)) + 18 \\ 2 = \frac{2\pi}{B} \rightarrow B = \pi, \quad c = \frac{1}{2} \\ \therefore h(t) = 15 \sin\left(\pi t - \frac{\pi}{2}\right) + 18 \end{cases}$$

using seconds

$$\begin{cases} h(t) = 15 \sin\left(\frac{\pi}{60}(t - c)\right) + 18 \\ 120 = \frac{2\pi}{B} \rightarrow B = \frac{\pi}{60} \\ c = 30 \\ \therefore h(t) = 15 \sin\left(\frac{\pi}{60}t - \frac{\pi}{2}\right) + 18 \end{cases}$$

- (d) Write down the height of the green light

(i) after 3 minutes: $h(3) = 33$ m (or $h(3 \cdot 60)$)

(ii) after 4.5 minutes: $h(4.5) = 18$ m (or $h(270)$)

(iii) after 5.15 minutes: $h(5.15) = 31.4$ m (or $h(309)$)